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```
clc;
clear;
close all;

format short g;
```

INPUT PARAMETERS

```
b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];

Bc = 8;                % Coefficient word length
Bp = 16;               % Product word length
N = 200;

x = [1 zeros(1,N-1)];
n = 0:N-1;
```

1. COEFFICIENT QUANTIZATION

```
Sc = 2^(Bc-1)-1;

bq = round(b*Sc)/Sc;
aq = round(a*Sc)/Sc;
```

2. IDEAL AND COEFFICIENT-QUANTIZED RESPONSES

```
y_ideal = filter(b,a,x);
y_coeff = filter(bq,aq,x);

e_coeff = y_ideal-y_coeff;

MaxCoeffError = max(abs(e_coeff));

SQNR_coeff = 10*log10( ...
    sum(y_ideal.^2)/sum(e_coeff.^2));
```

3. PRODUCT QUANTIZATION

```
% Quantize individual products to a signed fixed-point grid.
% Products are rounded before being accumulated.

Sp = 2^(Bp-1)-1;

y_prod = zeros(1,N);
xhist = zeros(1,length(bq));
yhist = zeros(1,length(aq)-1);

for k = 1:N

    % Current input and previous input samples
    xhist = [x(k) xhist(1:end-1)];

    % Feedforward products
    pf = bq .* xhist;

    % Feedback products
    pb = -aq(2:end) .* yhist;

    % Quantize each product
    pfq = round(pf*Sp)/Sp;
    pbq = round(pb*Sp)/Sp;

    % Accumulate quantized products
    y_prod(k) = sum(pfq)+sum(pbq);

    % Update previous output samples
    yhist = [y_prod(k) yhist(1:end-1)];

end

e_prod = y_coeff-y_prod;

MaxProdError = max(abs(e_prod));
```

```
SQNR_prod = 10*log10( ...
    sum(y_coeff.^2)/max(sum(e_prod.^2),eps));
```

4. ZERO-INPUT LIMIT-CYCLE TEST

```
x0 = zeros(1,N);

state = [0.01 0.01 0.01 0.01];
y_zero = zeros(1,N);

for k = 1:N

    % Feedback products for zero input
    pz = -aq(2:end).*state;

    % Product quantization
    pzq = round(pz*Sp)/Sp;

    % Zero-input output
    y_zero(k) = sum(pzq);

    % Update feedback states
    state = [y_zero(k) state(1:end-1)];

end

LimitAmplitude = max(abs(y_zero));

LimitSamples = sum(abs(y_zero)>1e-12);
```

5. OVERFLOW AND SATURATION TEST

```
xo = 1.2*ones(1,N);

y_over = filter(bq,aq,xo);

% Saturation to the range [-1,1]
y_sat = min(max(y_over,-1),1);

OverflowCount = sum(abs(y_over)>1);
MaxOverflow = max(abs(y_over));
```

6. DISPLAY NUMERICAL RESULTS

```
fprintf('\n=============================================\n');
fprintf('          QUANTIZATION RESULTS\n');
fprintf('=============================================\n');

fprintf('Maximum coefficient/output error = %.8f\n', ...
    MaxCoeffError);

fprintf('Coefficient SQNR = %.2f dB\n',SQNR_coeff);
```

```
fprintf('Maximum product quantization error = %.8f\n', ...
    MaxProdError);

fprintf('Product SQNR = %.2f dB\n',SQNR_prod);

fprintf('Zero-input maximum amplitude = %.9f\n', ...
    LimitAmplitude);

fprintf('Overflow samples = %d\n',OverflowCount);
```

```
==========================================
          QUANTIZATION RESULTS
==========================================
Maximum coefficient/output error = 0.00440576
Coefficient SQNR = 32.23 dB
Maximum product quantization error = 0.00021363
Product SQNR = 48.18 dB
Zero-input maximum amplitude = 0.009216590
Overflow samples = 194
```

7. DISPLAY ORIGINAL AND QUANTIZED COEFFICIENTS

```
fprintf('\nOriginal b =\n');
disp(b);

fprintf('Quantized b =\n');
disp(bq);

fprintf('Original a =\n');
disp(a);

fprintf('Quantized a =\n');
disp(aq);
```

```
Original b =
       0.0048       0.0193       0.0289       0.0193       0.0048

Quantized b =
     0.007874     0.015748     0.031496     0.015748     0.007874

Original a =
            1      -2.3695        2.314      -1.0547       0.1874

Quantized a =
            1      -2.3701        2.315      -1.0551      0.18898
```

8. RESULTS TABLE

```
Test = {
    'Coefficient Quantization'
    'Product Quantization'
    'Zero-Input Test'
    'Overflow Test'
};

MaximumError = [
    MaxCoeffError
    MaxProdError
    LimitAmplitude
    MaxOverflow
];

SQNR_dB = [
    SQNR_coeff
    SQNR_prod
    NaN
    NaN
];

Results = table(Test,MaximumError,SQNR_dB);

fprintf('\nRESULTS TABLE\n');
disp(Results);
```

```
RESULTS TABLE
                Test                MaximumError    SQNR_dB
    ____________________________    ____________    _______

    {'Coefficient Quantization'}      0.0044058     32.233
    {'Product Quantization'    }     0.00021363     48.176
    {'Zero-Input Test'         }      0.0092166        NaN
    {'Overflow Test'           }         1.3599        NaN
```

GRAPH 1: IDEAL VS COEFFICIENT-QUANTIZED RESPONSE

```
figure('Name','Ideal vs Coefficient-Quantized Response');

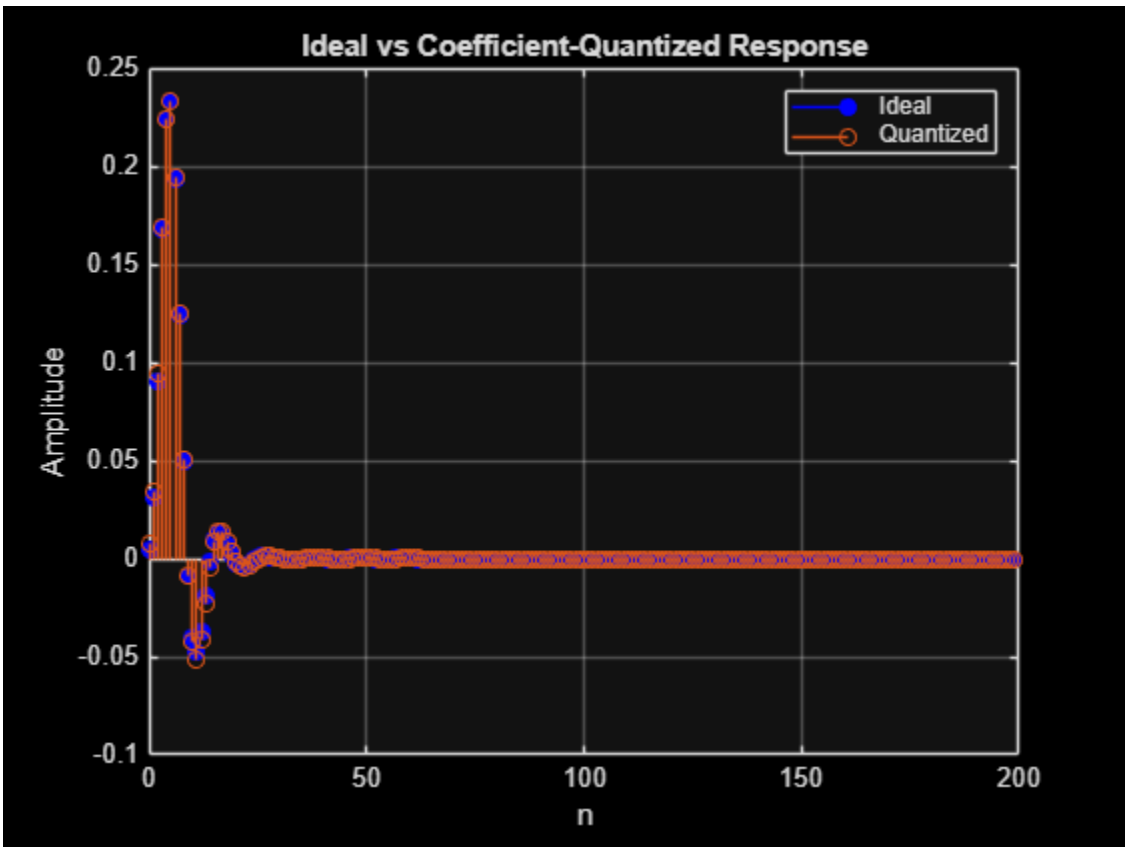
stem(n,y_ideal,'b','filled');
hold on;
stem(n,y_coeff,'o','Color',[0.85 0.33 0.10]);

grid on;
xlim([0 N]);
xlabel('n');
```

```
ylabel('Amplitude');

title('Ideal vs Coefficient-Quantized Response');

legend('Ideal','Quantized');
```



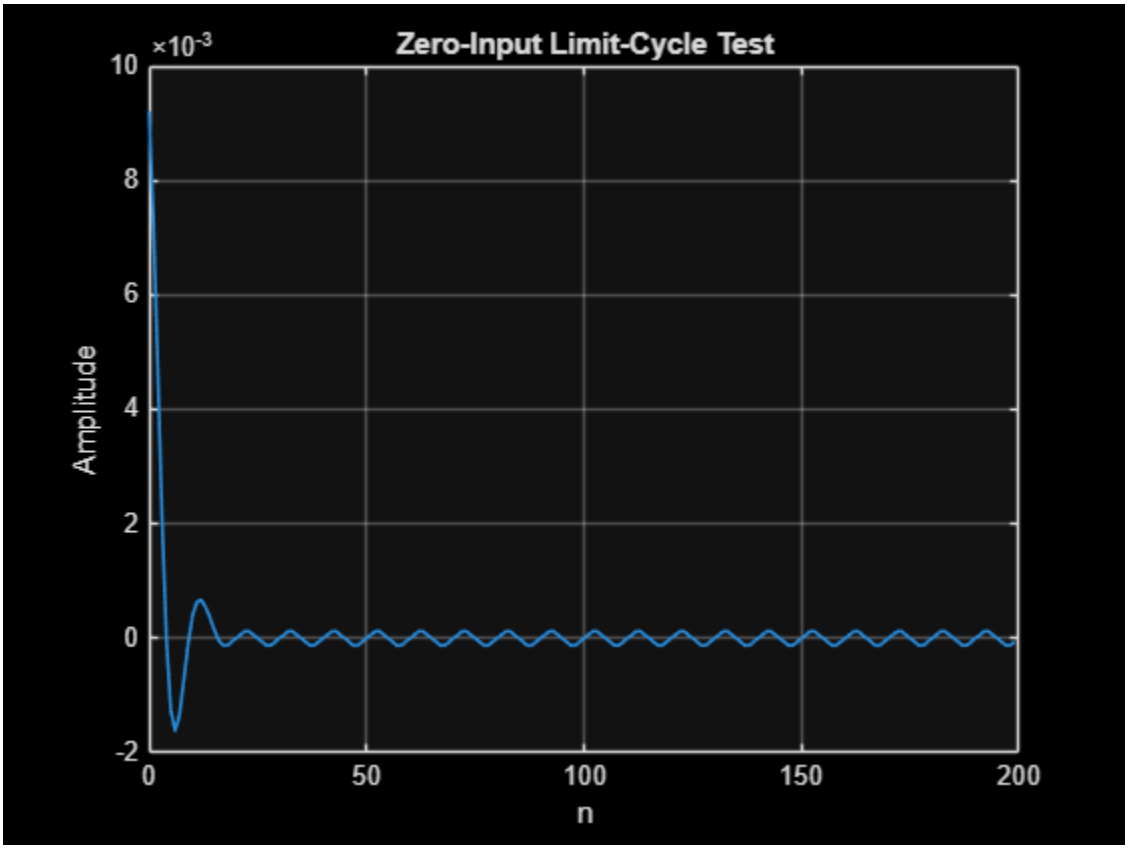
GRAPH 2: ZERO-INPUT LIMIT-CYCLE TEST

```
figure('Name','Zero-Input Limit-Cycle Test');

plot(n,y_zero,'LineWidth',1.2);

grid on;
xlim([0 N]);
xlabel('n');
ylabel('Amplitude');

title('Zero-Input Limit-Cycle Test');
```



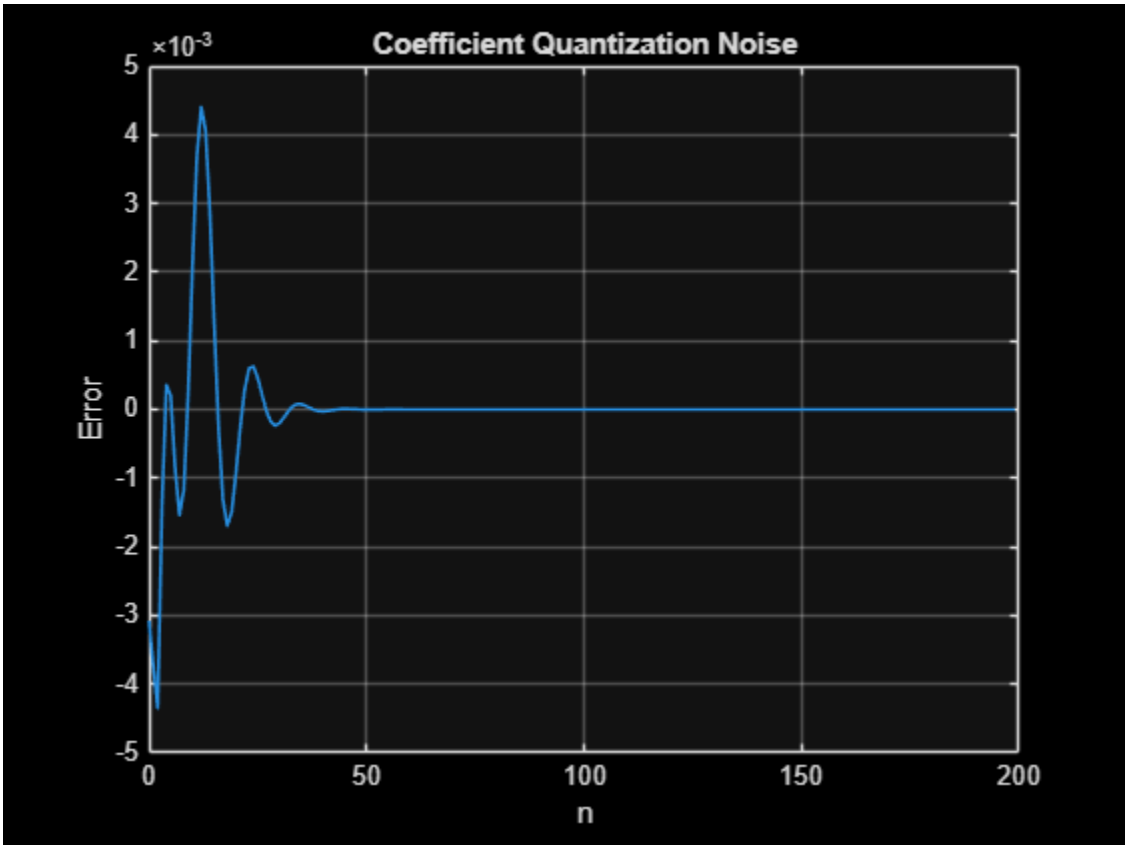
GRAPH 3: COEFFICIENT QUANTIZATION NOISE

```
figure('Name','Coefficient Quantization Noise');

plot(n,e_coeff,'LineWidth',1.2);

grid on;
xlim([0 N]);
xlabel('n');
ylabel('Error');

title('Coefficient Quantization Noise');
```



GRAPH 4: OVERFLOW AND SATURATION TEST

```
figure('Name','Overflow Saturation Test');

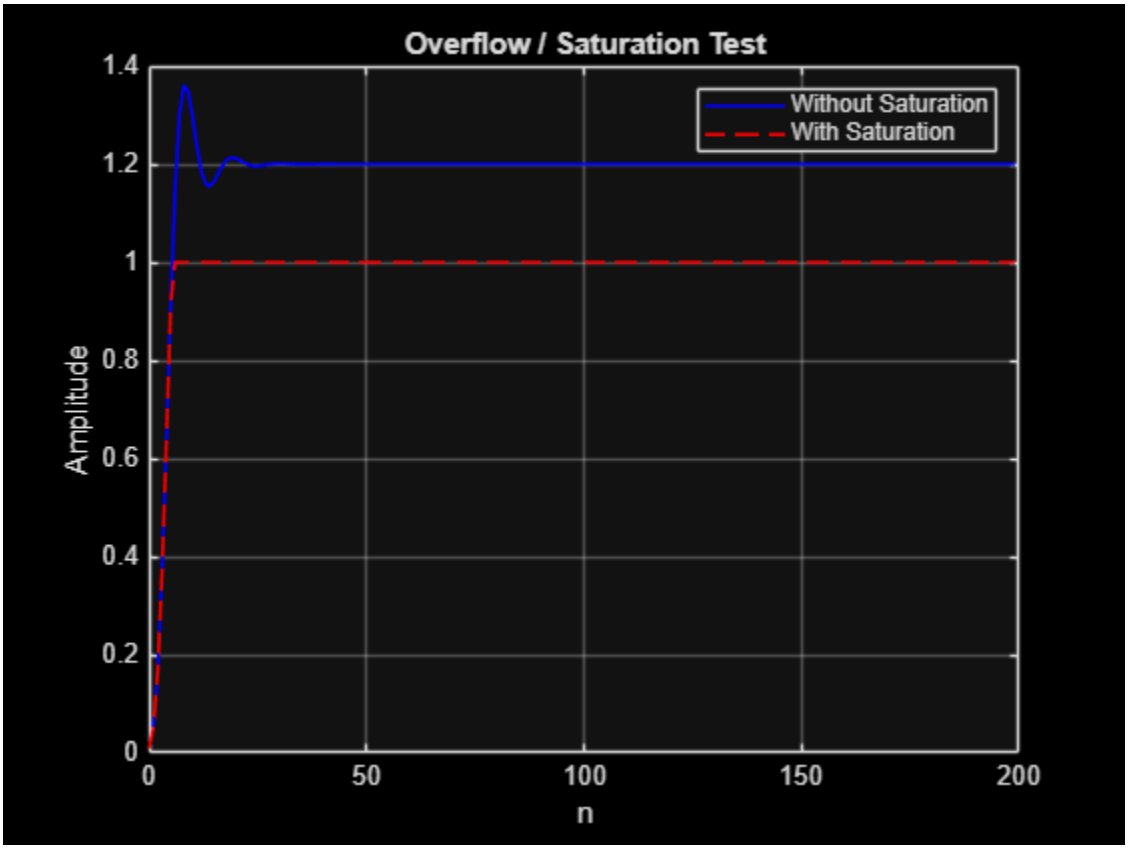
plot(n,y_over,'b','LineWidth',1.2);
hold on;

plot(n,y_sat,'r--','LineWidth',1.2);

grid on;
xlim([0 N]);
xlabel('n');
ylabel('Amplitude');

title('Overflow / Saturation Test');

legend('Without Saturation','With Saturation');
```

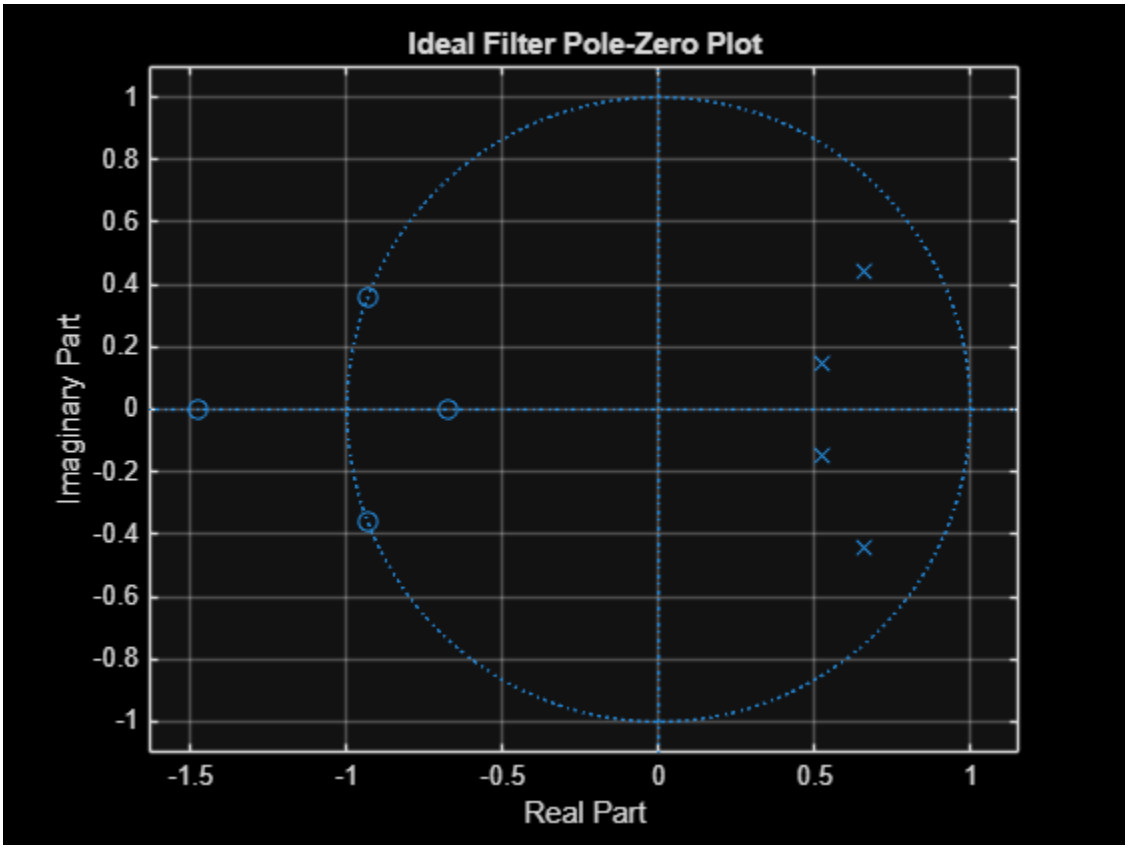


GRAPH 5: IDEAL FILTER POLE-ZERO PLOT

```
figure('Name','Ideal Filter Pole-Zero Plot');

zplane(b,a);
grid on;

title('Ideal Filter Pole-Zero Plot');
```

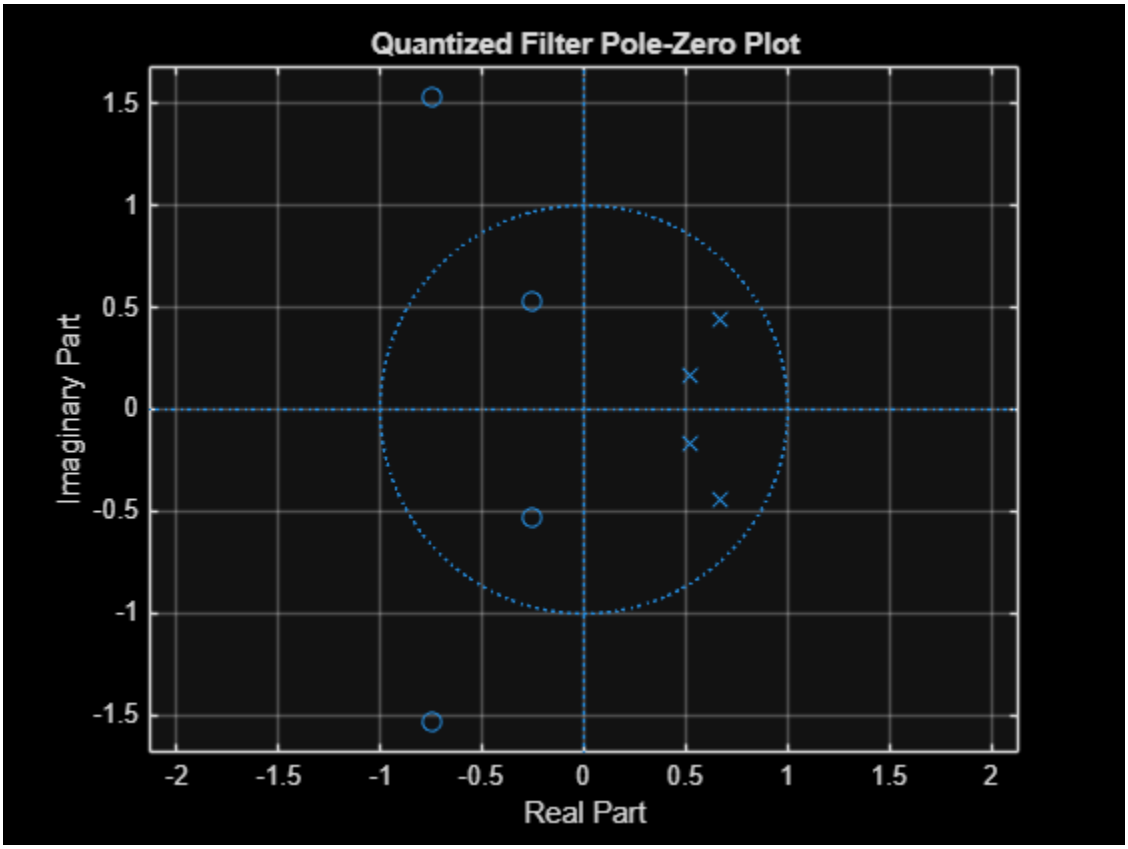


GRAPH 6: QUANTIZED FILTER POLE-ZERO PLOT

```
figure('Name','Quantized Filter Pole-Zero Plot');

zplane(bq,aq);
grid on;

title('Quantized Filter Pole-Zero Plot');
```



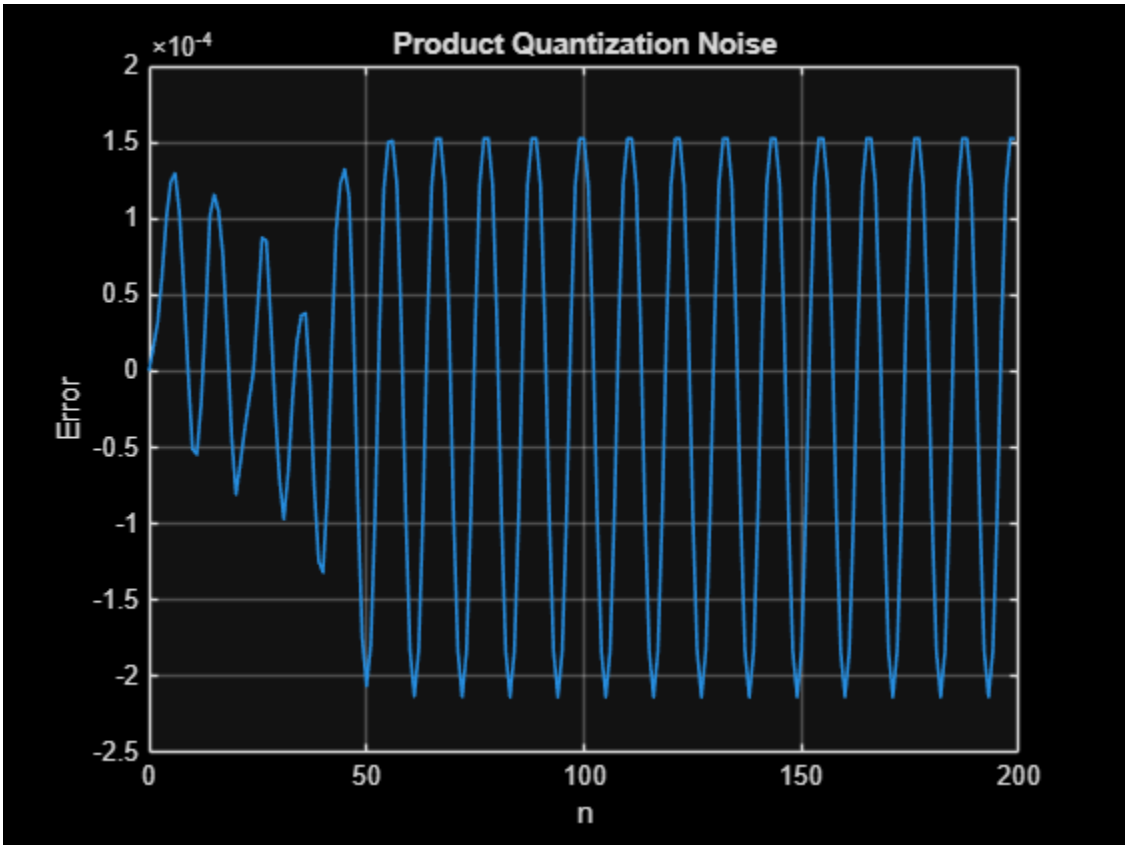
GRAPH 7: PRODUCT QUANTIZATION NOISE

```
figure('Name','Product Quantization Noise');

plot(n,e_prod,'LineWidth',1.2);

grid on;
xlim([0 N]);
xlabel('n');
ylabel('Error');

title('Product Quantization Noise');
```



FINAL RESULT

```
fprintf('\n==========================================\n');
fprintf('              FINAL RESULT\n');
fprintf('==========================================\n');

fprintf('1. Coefficient quantization was performed.\n');
fprintf('2. Product quantization noise was calculated.\n');
fprintf('3. Zero-input response was investigated.\n');
fprintf('4. Overflow and saturation were investigated.\n');
fprintf('5. Ideal and quantized pole-zero plots were generated.\n');
fprintf('6. Quantization effects were analyzed.\n');
```

```
==========================================
              FINAL RESULT
==========================================
1. Coefficient quantization was performed.
2. Product quantization noise was calculated.
3. Zero-input response was investigated.
4. Overflow and saturation were investigated.
5. Ideal and quantized pole-zero plots were generated.
6. Quantization effects were analyzed.
```

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